

POWERFACTORY

PowerFactory 2024

Technical Reference

Multi-Winding Transformer

ElmTrmult,TypTrmult

PF2024

POWER SYSTEM SOLUTIONS

MADE IN GERMANY

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1 General Description

The multi-winding transformer model in *PowerFactory* allows to model transformers with 5 up to 15 windings. The multi-winding transformer model is presented in this document.

A schematic representation of the multi-winding transformer is shown in Figure 1.1.

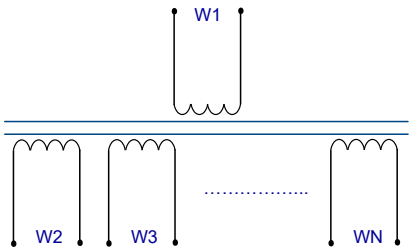


Figure 1.1: Schematic representation of a multi-winding transformer. The number of windings *N* can be between 5 and 15.

The multi-winding transformer with *N* windings is an *N*-port element connecting *N* cubicles in the network. The number of windings (*N*) can be between 5 and 15. The transformer model in *PowerFactory* consists of the multi-winding transformer element (*ElmTrmult*) and the multi-winding transformer type (*TypTrmult*).

The *N* windings are referred to as winding W1, W2, ..., WN.

The multi-winding transformer model supports *D*, *Y*, *YN*, *Z* and *ZN* vector groups. Grounding of the star-points can be defined in the element. Neutral connection is not supported by the multi-winding transformer model.

The multi-winding transformer model supports only one tap changer, which can be located at any of the windings. The tap model is equivalent to the tap model of the three-phase 2-winding transformer model. However, the multi-winding transformer model does not support the definition of the external tap controller, the external substation controller and the internal (LDC) line drop compensation.

Saturation of the magnetizing reactance is represented in the same way as for the three-phase 2-winding transformer model. The magnetizing impedance can be located at the terminals connection of any of the *N* windings.

Please refer to the three-phase 2-winding transformer technical reference for further details about the tap and saturation modelling.

When drawing the multi-winding transformer element in the grid, the number of winding is automatically requested in order to draw the correct symbol. The number of windings defined for the element must be the same as the number of windings defined in the type.

2 Modelling

Modelling of the multi-winding transformer is based on the theory presented in [1], which represents the transformer by an admittance matrix formulation. It is noted that the multi-winding transformer model is not based on an explicit equivalent circuit, as done for the four-winding transformer. The magnetizing impedance is always modelled at a terminal and it is not included in the admittance matrix.

2.1 Fundamental frequency calculations

2.1.1 Positive- and negative-sequence model

At the positive- and negative-sequence, the *N*-winding transformer is represented by an *NxN* admittance matrix. The user can select at which terminal the magnetizing impedance (comprising the magnetising reactance and iron losses) is connected. The transformer taps are always modelled at the terminals of the selected winding.

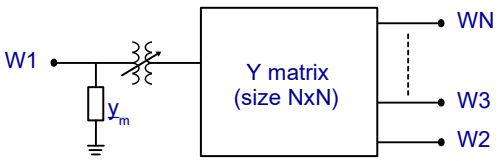


Figure 2.1: Positive-sequence representation of the *N*-winding transformer with both tap and magnetizing impedance modelled at winding W1.

Input data for the positive- and negative-sequence model are the winding resistances and leakage reactances. The admittance matrix is built starting from the winding resistances and leakage reactances data, according to the theory presented in [1].

If the complete input data option is selected, it is possible to enter the leakage reactance for each pair of windings. Totally, $N \cdot (N-1)/2$ leakage reactance values must be entered in p.u. based on the minimum rated power of the two considered windings.

If all detailed measurements between each pair of windings are not available, it is possible to enter the leakage reactance data in a simplified way, i.e. the user can select the simplified input data option and define one reactance value for each winding (totally *N* values). Each reactance must be entered in p.u. based on the rated power of the corresponding winding. If the simplified input data option is selected, the admittance matrix is built under the assumption that the short-circuit reactance between any pair of windings is given by the sum of the two reactances of the two windings. In this case, it is possible to draw an equivalent star circuit for the multi-winding transformer. This is a simplification and this circuit is not able in general to reproduce all short-circuit impedance measurements between each pair of windings.

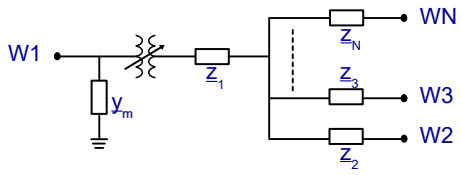


Figure 2.2: Positive-sequence simplified equivalent circuit of the *N*-winding transformer with both tap and magnetizing impedance modelled at winding W1.

Note that the winding resistances (copper losses) are entered per each winding also when the complete input data option has been selected. Therefore, always *N* resistance values are required.

The magnetising admittance $\underline{y_m}$ (parallel representation of conductance and susceptance) is calculated as:

$$y_m = \frac{curmag}{100} \quad \text{p.u.} \tag{1}$$

$$\underline{y_m}.r = \frac{pfe}{1000} / S_{r_W1} \quad \text{p.u.} \tag{2}$$

$$\underline{y_m}.i = -\sqrt{y_m^2 - \underline{y_m}.r^2} \quad \text{p.u.} \tag{3}$$

where pfe are the measured no-load losses in kW and $curmag$ is the no-load current in %.

If the tap is located at winding *k*, the complex winding ratio is calculated according to Equation 4:

$$\underline{t_k} = (1 + t_{dutap} \cdot \underline{t_{phit}}) \cdot \underline{t_{trangle_k}} \tag{4}$$

where:

$$t_{dutap} = (ntap - ntapneu) \cdot \frac{dutap}{100} \tag{5}$$

$$\underline{t_{phitr}} = \cos\left(\frac{phitr \cdot \pi}{180}\right) + \jmath \cdot \sin\left(\frac{phitr \cdot \pi}{180}\right) \tag{6}$$

$$\underline{t_{trangle_k}} = \cos\left(\frac{trangle_k \cdot \pi}{6}\right) - \jmath \cdot \sin\left(\frac{trangle_k \cdot \pi}{6}\right) \tag{7}$$

The parameter $ntap$ is the actual tap position (element parameter), $ntapneu$ neutral tap position (type parameter), $dutap$ additional voltage per tap (type parameter), $phitr$ phase of du (type parameter) and $trangle_k$ is the phase shift of winding Wk (type parameter). It is noted that $trangle_k$ is the element in the *kth* row of the *Phase shift* column in the *Winding configuration* table on the Basic Data page of the type.

2.1.2 Zero-sequence model

At the zero-sequence, the *N*-winding transformer is represented by an $N0xN0$ admittance matrix, where $N0$ is equal to the number of windings with vector group different than *Z* or *ZN*. *YN*, *Y* and *D* windings are included in the admittance matrix. Ungrounded *Y* windings are treated internally as if grounded through very large impedance. *D* windings are considered as short-circuited windings (at the zero-sequence). *Z* and *ZN* windings are modelled with separate equations as grounded impedances. Additionally, for each *D* winding, the zero-sequence grid current is set to zero. The zero-sequence representation of the multi-winding transformer is shown in Figure 2.3.

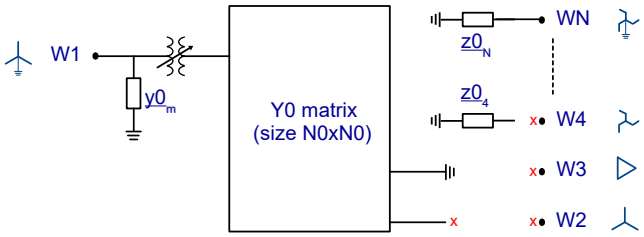


Figure 2.3: Zero-sequence representation of the *N*-winding transformer with both tap and magnetizing impedance modelled at winding W1. *N0* is the number of windings with vector group different than *Z* and *ZN*. The figure shows how each vector group is treated internally at the zero-sequence. Any vector group can be assigned at each winding.

The zero-sequence magnetizing impedance is located at the same location as the positive-sequence magnetizing impedance. The zero-sequence magnetizing impedance is always modelled at a terminal and it is not included in the admittance matrix. If the magnetization is located at a *ZN* winding or if there is at least one *D* winding, the zero-sequence magnetization is ignored.

The zero-sequence magnetising admittance $\underline{y_{m0}}$ is calculated using the zero-sequence no-load current $cur0mag$ and the zero-sequence r to x magnitude $rtox0_n$ as:

$$\underline{z_{m0}}.i = \frac{100}{cur0mag} / \sqrt{1 + rtox0_n^2} \quad \text{p.u.} \tag{8}$$

$$\underline{z_{m0}}.r = \underline{z_{m0}}.i \cdot rtox0_n \quad \text{p.u.} \tag{9}$$

$$\underline{y_{m0}} = 1/\underline{z_{m0}} \quad \text{p.u.} \tag{10}$$

Input data for the zero-sequence model are the winding zero-sequence resistances and leakage reactances.

If the complete input data option is selected, it is possible to enter the leakage reactance for each pair of windings. Totally, $N \cdot (N-1)/2$ leakage reactance values must be entered in p.u. based on the minimum rated power of the two considered windings.

If all detailed measurements between each pair of windings are not available, it is possible to enter the leakage impedance data in a simplified way, i.e. the user can select the simplified input data option and define one reactance value for each winding (totally *N* values). Each reactance must be entered in p.u. based on the rated power of the corresponding winding. If the simplified input data option is selected, the admittance matrix is built under the assumption that the short-circuit reactance between any pair of windings is given by the sum of two reactances of the two windings.

If at least one *Z* or *ZN* winding is defined, only the simplified input data mode for the zero-sequence reactances is supported.

Note that the winding resistances (copper losses) are entered per each winding also when the complete input data option has been selected. Therefore, always *N* resistance values are required.

2.2 EMT simulation model

The EMT multi-winding transformer model is based on the reduced impedance matrix approach presented in [1]. Modelling is performed in the alpha-beta-gamma components. At the gamma components, *Z* and *ZN* windings are modelled with separate equations, *D* windings are considered internally as short-circuited windings (as already described in 2.1.2).

Modelling of saturation, including hysteresis, is supported for the magnetizing reactance in the same way as for the two-winding transformer.

The magnetising conductance and susceptance (in series for the EMT model) are calculated using the no-load current $curmag$ and no-load losses pfe as:

$$y_m = \frac{curmag}{100} \quad \text{p.u.} \tag{11}$$

$$g_m = \frac{pfe}{1000} / S_{r_W1} \quad \text{p.u.} \tag{12}$$

$$b_m = \sqrt{y_m^2 - g_m^2} \cdot (2 \cdot \pi \cdot f_{nom}) \quad \text{p.u.} \tag{13}$$

The zero-sequence values are obtained as:

$$x_{m0} = \frac{100}{cur0mag} / \sqrt{1 + rtox0_n^2} \tag{14}$$

$$r_{m0} = rtox0_n \cdot x_{m0} \tag{15}$$

$$b_{m0} = \frac{2 \cdot \pi \cdot f_{nom}}{x_{m0}} \tag{16}$$

3 Parameter description

3.1 Access to winding parameters

There are a number of array parameters of length *N* which refer to each winding. Access to the *kth* value, referring to winding *k*, is possible in the Attribute selection *IntMon* by choosing the corresponding subindex or directly in the Attribute selection editor by adding *:k* to the name of the parameter (where *k* is between 0 and 14). Examples of such array parameters are the loading *loading_w*, the nominal current *Inom_w*, the rating factor *ratfac*, the star-point voltage *Ustar*, the star-point current *Istar*, the zero-sequence circulating current in a delta-winding *i0Delta* and *I0Delta*.

3.2 Relationship between per-unit and absolute values

3.2.1 Reactance between two windings

A leakage reactance in Ω between a pair of windings *i* and *j*, as seen from winding *j* and referred to the voltage of winding *j*, needs to be divided by the reference impedance between the two windings to get the p.u. value:

$$x_{ij} = \frac{X_{\Omega,ij}}{Z_{ref,ij}} \quad \text{p.u.} \tag{17}$$

The reference impedance is calculated as:

$$Z_{ref,ij} = \frac{U_{r,j}^2}{S_{r_min}} \qquad \Omega \tag{18}$$

where $U_{r,j}$ is the rated voltage of winding *j* in kV and S_{r_min} is the minimum of the rated powers between the two windings.

For example, the following is valid for the $X_{\Omega,W1-W2}$ impedance given in Ω referred to the rated voltage of the W2 winding:

$$x_{W1-W2} = X_{\Omega,W1-W2} \cdot \frac{min(S_{r,W1},S_{r,W2})}{U_{r,W2}^2} \qquad \text{p.u.} \tag{19}$$

3.2.2 Reactance of a winding

In the case of the simplified input mode, the p.u. reactance of winding *i*, given its value in Ω referred to the winding nominal voltage, is calculated as:

$$x_i = \frac{X_{\Omega,i}}{Z_{ref,i}} \qquad \text{p.u.} \tag{20}$$

The reference impedance is calculated as:

$$Z_{ref,i} = \frac{U_{r,i}^2}{S_{r,i}} \qquad \Omega \tag{21}$$

where $U_{r,i}$ and $S_{r,i}$ are the rated voltage and power of winding *i* in kV and MVA respectively.

For example, the following is valid for the $X_{\Omega,W2}$ impedance given in Ω referred to the rated voltage of the W2 winding:

$$x_{W2} = X_{\Omega,W2} \cdot \frac{S_{r,W2}}{U_{r,W2}^2} \qquad \text{p.u.} \tag{22}$$

3.2.3 Resistance of a winding

The p.u. resistance of winding *i*, given its value in Ω referred to the winding nominal voltage, is calculated as:

$$r_i = \frac{R_{\Omega,i}}{Z_{ref,i}} \qquad \text{p.u.} \tag{23}$$

where $Z_{ref,i}$ is given in equation 21.

For example, the following is valid for the $R_{\Omega,W2}$ impedance given in Ω referred to the rated voltage of the W2 winding:

$$r_{W2} = R_{\Omega,W2} \cdot \frac{S_{r,W2}}{U_{r,W2}^2} \qquad \text{p.u.} \tag{24}$$

3.3 No-load current

If the no-load current is measured from e.g. the second winding W2, the no-load current $curmag$ in %, based on the apparent power of the first winding W1, is calculated from the measured no-load current I_0 in kA at the W2 winding as:

$$curmag = \frac{I_0}{I_{r,W2}} \cdot \frac{S_{r,W2}}{S_{r,W1}} \cdot 100 \qquad where \qquad I_{r,W2} = \frac{S_{r,W2}}{\sqrt{3} \cdot U_{r,W2}} \tag{25}$$

where $S_{r,W2}$ is the rated apparent power in MVA of the W2 winding, $S_{r,W1}$ is the rated apparent power in MVA of the first winding W1 and $U_{r,W2}$ is the rated voltage in kV of the W2 winding. Similar formulas apply if the magnetization current is measured from a different winding than W2.

3.4 Nominal power and current

The multi-winding transformer model in *PowerFactory* does not support the definition of a characteristic for the rating factor, nor it supports the definition of a thermal rating object.

The calculation parameter $Inom_w$ (array of length *N*) takes into consideration the actual tap position, depending on the tap dependent rating configuration, see section 3.4.1.

3.4.1 Nominal current

The calculation of the nominal current of a winding with a tap defined is tap dependent, according with the selected tap configuration. The nominal currents of all other windings is not tap dependent. The winding nominal currents are available in the array calculation parameter $Inom_w$.

The nominal current for winding *k* is determined as follows:

$$Inom_w_k = ratfac_k \cdot I_{r,k,tap} \cdot ntnum \tag{26}$$

where

- $ratfac_k$ is the rating factor for winding *k*, defined on the element *Basic Data* page
- $ntnum$ is the number of parallel transformers, defined on the element *Basic Data* page
- $I_{r,k,tap}$ is described below

The tap dependent rated current $I_{r,k,tap}$ is calculated, only for the winding with a tap defined, depending on the type parameter *iTaploading (Tap dependent rating - Configuration)*:

- *No dependency (Constant current)*
  The nominal current is constant and independent of the tap position:

$$I_{r,k,tap} = \frac{S_{r,k}}{\sqrt{3} \cdot U_{r,k}} \quad \text{kA} \tag{27}$$

- *Full Capacity Below Normal (FCBN)*
  The nominal current is decreased for taps resulting in voltage above normal and increased for taps resulting in voltage below normal. If supplied with the voltage corresponding to the actual tap position, the transformer will supply full capacity (rated MVA) at each tap position:

$$I_{r,k,tap} = \frac{S_{r,k}}{\sqrt{3} \cdot U_{r,k} \cdot |\underline{tapRatio}|} \quad \text{kA} \tag{28}$$

- *Reduced Capacity Below Normal (RCBN)*
  The nominal current is decreased for taps resulting in voltage above normal. It is constant and equal to rated current for taps resulting in voltage below normal. If supplied with the voltage corresponding to the actual tap position, the transformer will supply full capacity (rated MVA) for taps resulting in voltage above normal, but reduced capacity for taps resulting in voltage below normal:
  - tap position *above* neutral tap position:

$$I_{r,k,tap} = \frac{S_{r,k}}{\sqrt{3} \cdot U_{r,k} \cdot |\underline{tapRatio}|} \quad \text{kA} \tag{29}$$

  - tap position *below* neutral tap position:

$$I_{r,k,tap} = \frac{S_{r,k}}{\sqrt{3} \cdot U_{r,k}} \quad \text{kA} \tag{30}$$

where

$$\underline{tapRatio} = 1 + (tap - nntap0) \cdot \frac{dutap}{100} \cdot \left(cos(phitr) + j \cdot sin(phitr) \right) \tag{31}$$

and

- $nntap$ is the neutral tap position defined in the type
- $dutap$ is the additional voltage per tap defined in the type
- $phitr$ is the phase of dutap defined in the type
- tap is the actual tap position in the calculation or simulation
- $S_{r,k}$ is the rated power in MVA for the k-side
- $U_{r,k}$ is the k-winding rated voltage in kV

For all other winding without a tap, then the corresponding of equations 30 is always used at that side.

4 Load Flow Analysis

4.1 Operational limits for tap changer

Operational limits for the tap can be optionally defined in the transformer element *Load Flow* page. A characteristic can be applied to the operational limits. The operational tap limits must be within the minimum and maximum tap positions defined in the transformer type, otherwise a warning message is printed in the output window.

Before running a load flow, it is checked that the actual tap position is within the minimum and maximum tap positions defined in the type and within the operation limits. If the initial tap position is outside the type limits, it is forced to the limit and a warning message is printed in the output window. If the initial tap position is outside the operational limits a warning message is printed in the output window, but the initial tap position is not changed as long as it is within the type limits.

During the load flow calculation, the actual tap position is constrained within the operational limits if defined, otherwise it is constrained within the type limits for the tap.

4.2 Calculation Quantities for AC Load Flow

4.2.1 Loading

The loading of the transformer is calculated as follows:

$$loading_w_k = \frac{I_k}{Inom_w_k} \cdot 100\ \%$$
$$loading = max(loading_w_1, loading_w_2, ..., loading_w_N) \quad \%$$

- $loading$: Maximum loading in %
- $loading_w_k$: Loading of winding *k* in %
- $Inom_w_k$: Nominal current of winding *k* in kA, see section 3.4.1
- I_k : Magnitude of the current at winding *k* in kA

For an unbalanced load flow calculation the highest current of all phases/neutral is used.

4.2.2 Losses

For an *N* winding transformer, the losses are calculated as follows:

Table 4.1: Losses Quantities

Quantity	Unit	Description	Value
$Ploss$	MW	Losses (total)	$= \sum_{n=1}^{N} P_k$
$Qloss$	Mvar	Reactive-Losses (total)	$= \sum_{n=1}^{N} Q_k$
$Plossld$	MW	Losses (load)	$= Ploss - Plossnld$
$Qlossld$	Mvar	Reactive-Losses (load)	$= Qloss - Qlossnld$
$Plossnld$	MW	Losses (no load)	$Gmload/1000$
$Qlossnld$	Mvar	Reactive-Losses (no load)	$Xmload/1000$

where $Gmload$ and $Xmload$ are calculated as:

$$Gmload = Re(\underline{u_{mag}} \cdot \underline{i_{mag}}^*) \cdot S_{r,1} \cdot ntnum \cdot 1000 \quad \text{kW}$$
$$Xmload = Im(\underline{u_{mag}} \cdot \underline{i_{mag}}^*) \cdot S_{r,1} \cdot ntnum \cdot 1000 \quad \text{kvar}$$

and

- $\underline{u_{mag}}$ is the actual phasor voltage over the transformer magnetising branch in p.u.
- $\underline{i_{mag}}$ is the actual phasor current over the transformer magnetising branch in p.u.
- $S_{r,1}$ is the rated power of winding W1
- P_k is the total active power at winding *k* in MW
- Q_k is the total reactive power at winding *k* in Mvar
- $ntnum$ is the number of parallel transformers

4.3 Calculation Quantities for linear DC Load Flow

4.3.1 Loading

For an *N* winding transformer, the loading of the transformer is calculated as follows:

$$loading_w_k = \frac{|P_k|}{Pnom_k} \cdot 100\ \%$$
$$loading = max(loading_w_1, loading_w_2, ..., loading_w_N) \quad \%$$

where:

- P_k : Active power at terminal *k*
- $Pnom_k$: Nominal power at side *k*

The the nominal power is calculated as:

- $Pnom_k = \sqrt{3} \cdot U_{n,k} \cdot Inom_w_k$

with the nominal current $Inom_w_k$ calculated according to equations 26 and 30. $U_{n,k}$ is the busbar line-line nominal voltage in kV at the side *k*.

4.3.2 Losses

Losses are not calculated in the linear DC Load Flow.

5 Harmonics/Power Quality

On the *Harmonics* page of the transformer type, frequency-dependent characteristics can be defined for the positive-sequence resistances and reactances. The characteristic defined for $r1(f)$ is applied to all positive-sequence winding resistances. The characteristic defined for $l1(f)$ is applied to all positive-sequence winding inductances.

Frequency-dependent characteristics can be defined also for the zero-sequence resistances and reactances. The characteristic defined for $r0(f)$ is applied to all zero-sequence winding resistances. The characteristic defined for $l0(f)$ is applied to all zero-sequence winding inductances.

Therefore, all positive-sequence resistances are assumed to have the same frequency dependency (according to $r1(f)$). The same is true for the positive-sequence inductances (according to $l1(f)$), zero-sequence resistances (according to $r0(f)$) and inductances (according to $l0(f)$).

A frequency-dependent positive-sequence magnetizing admittance can also be defined, by defining a characteristic for the imaginary and real part of the admittance.

If the characteristic is defined through a *Frequency Polynomial Characteristic (ChaPol)* element, then it is automatically defined as relative to rated values. If the characteristic is defined through a *Parameter Characteristic - Vector (ChaVec)* element, then it can be defined as relative (in p.u. or %) or in absolute p.u. values.

6 Inputs variables of the load flow model

Table 6.1: Input definition of the load flow model

Input Signal	Symbol	Description	Unit
nt_in		Tap position, controller input	

7 Inputs/Outputs/State variables of the dynamic models

7.1 Stability Model (RMS)

Table 7.1: Input definition of the RMS-model

Input Signal	Symbol	Description	Unit
nntapin		Tap position, controller input	

Table 7.2: Output definition of the RMS-model

Parameter	Symbol	Description	Unit
I0rDelta		Circulating current in Delta-Winding, Real Part	kA
I0iDelta		Circulating current in Delta-Winding, Imaginary Part	kA

7.2 EMT-Model

Table 7.3: Input definition of the EMT-model

Input Signal	Symbol	Description	Unit
nntapin		Tap position, controller input	

Table 7.4: State variables definition of the EMT-model

Parameter	Symbol	Description	Unit
i0Delta		Zero sequence current in Delta winding (HV)	p.u.
psim_r		Magnetising flux alpha-component	p.u.
psim_i		Magnetising flux beta-component	p.u.
psim_0		Magnetising flux gamma-component	p.u.

Table 7.5: Output definition of the EMT-model

Parameter	Symbol	Description	Unit
psim_a		Magnetising flux—Mag. flux, phase a	p.u.
psim_b		Magnetising flux—Mag. flux, phase b	p.u.
psim_c		Magnetising flux—Mag. flux, phase c	p.u.

8 References

[1] V. Brandwajn, H. W. Dommel, and I. I. Dommel. Matrix representation of three-phase n-winding transformers for steady-state and transient studies. *IEEE Transactions on Power Apparatus and Systems*, 101(6):1369–1378, 1982.

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7.5 Output definition of the EMT-model 14